

lecture_20

September 19, 2026

1 Lecture 20: Duality — Non-Linear Programming

{note} Lectures 16-18 developed three algorithms - Penalty, Barrier, and Interior-Point - each of which computes a KKT point and, as a by-product, recovers a vector of Lagrange multipliers λ^* . Lecture 19 showed that this multiplier is already the NLP shadow price: $\lambda_i^* = -\partial f^* / \partial b_i$, the rate at which the optimal cost changes as a resource is relaxed. What Lecture 19 did not explain is *why* this is true - the result was stated and verified, but not derived from first principles. In Lecture 9 we showed that LP shadow prices are optimal dual variables - the solution to a second, mirror LP derived mechanically from the primal. This lecture develops the NLP analogue: the Lagrangian dual, a concave maximization problem whose optimal value lower-bounds the primal and, under a mild regularity condition, equals it exactly. The LP duality theorem from Lecture 9 falls out as a special case, and Lecture 19's Sensitivity Theorem falls out as a direct corollary of strong duality.

Learning Objectives

By the end of this notebook, you will be able to: 1. Construct the Lagrangian dual function and dual problem for a constrained NLP, and explain why the dual function is always concave. 2. State and apply the weak and strong duality theorems for convex NLPs, and recover LP duality (Lecture 9) as a special case. 3. Derive the NLP Sensitivity Theorem (Lecture 19) from strong duality, and verify by hand that the KKT multiplier λ^* simultaneously solves the dual problem and equals the shadow price, for transportation optimization problems.

Prerequisites: LP Duality (Lecture 9); NLP Principles (Lecture 12); Penalty, Barrier, and Interior-Point Methods (Lectures 16-18); NLP Sensitivity Analysis (Lecture 19).

Estimated time: 90 minutes (including in-class exercises)

1.1 Optimality Conditions — KKT

Consider the standard-form NLP:

$$\min_x f(x) \quad \text{subject to} \quad g_i(x) \leq 0, \quad i = 1, \dots, m$$

where $f, g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ are continuously differentiable. The Lagrangian is:

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{i=1}^m \lambda_i g_i(x)$$

If x^* is a local minimum and a constraint qualification holds (e.g., LICQ), then there exist multipliers $\lambda^* \in \mathbb{R}^m$ such that the following Karush–Kuhn–Tucker (KKT) conditions are satisfied:

Condition	Statement	Interpretation
Stationarity	$\nabla f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(x^*) = \mathbf{0}$	Gradient of objective balanced by gradients of active constraints
Primal feasibility	$g_i(x^*) \leq 0 \quad \forall i$	Solution lies within the feasible region
Dual feasibility	$\lambda_i^* \geq 0 \quad \forall i$	Multipliers are non-negative for inequality constraints
Complementary slackness	$\lambda_i^* g_i(x^*) = 0 \quad \forall i$	A constraint is either active ($g_i = 0$) or its multiplier is zero ($\lambda_i = 0$)

For a convex NLP (convex f , convex g_i), the KKT conditions are both necessary and sufficient for global optimality. All methods in Lectures 16–18 seek a point satisfying these four conditions — and in doing so, they recover the multipliers λ^* . This lecture shows that λ^* is precisely the optimal solution of the Lagrangian dual problem, and that $f(x^*) = q(\lambda^*)$ — strong duality.

1.2 NLP Duality

1.2.1 The Lagrangian Dual Function

Fix a multiplier vector $\lambda \geq \mathbf{0}$ and minimize the Lagrangian over all $x \in \mathbb{R}^n$ — with no constraint on x at all:

$$q(\lambda) = \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda) = \inf_{x \in \mathbb{R}^n} \left[f(x) + \sum_{i=1}^m \lambda_i g_i(x) \right]$$

The function $q : \mathbb{R}^m \rightarrow \mathbb{R} \cup \{-\infty\}$ is called the Lagrangian dual function. Three properties follow immediately:

Property 1 — q is always concave. $q(\lambda)$ is the pointwise infimum of a family of affine functions of λ (one for each fixed x). The infimum of affine functions is always concave, regardless of whether f or g_i are convex. This means the dual problem is always a concave maximization — and therefore has no local optima that are not global.

Property 2 — q lower-bounds the primal. For any feasible x (i.e., $g_i(x) \leq 0$ for all i) and any $\lambda \geq \mathbf{0}$:

$$\begin{aligned} q(\lambda) &\leq \mathcal{L}(x, \lambda) = f(x) + \underbrace{\sum_{i=1}^m \lambda_i g_i(x)}_{\leq 0 \text{ (since } \lambda_i \geq 0, g_i(x) \leq 0)} \leq f(x) \end{aligned}$$

So $q(\lambda) \leq f(x)$ for every feasible x and every $\lambda \geq \mathbf{0}$. Taking the infimum over x on the right gives: $q(\lambda) \leq f^*$ for all $\lambda \geq \mathbf{0}$, where $f^* = f(x^*)$ is the primal optimal value. Every dual-feasible λ provides a valid lower bound on the primal optimal.

Property 3 — If for some direction in $\mathbb{R}^n$ the Lagrangian is unbounded below, $q(\lambda) = -\infty$. Dual-feasible multipliers are those for which $q(\lambda) > -\infty$; the set of such λ is the dual feasible region $\mathcal{D} = \{\lambda \geq \mathbf{0} : q(\lambda) > -\infty\}$.

1.2.2 The Dual Problem

Since $q(\lambda)$ is a lower bound on f^* for every $\lambda \in \mathcal{D}$, the tightest such bound is the Lagrangian dual problem:

$$\max_{\lambda \geq \mathbf{0}} q(\lambda)$$

Let $q^* = \max_{\lambda \geq \mathbf{0}} q(\lambda)$ denote the dual optimal value, attained at some λ^* .

1.2.3 Weak Duality

For any primal-feasible x and any $\lambda \in \mathcal{D}$:

$$q(\lambda) \leq f^* \leq f(x)$$

In particular, the dual optimal value never exceeds the primal optimal value: $q^* \leq f^*$.

The gap $f^* - q^* \geq 0$ is the duality gap. Weak duality holds for *all* optimization problems — convex or not, linear or not — as long as $\lambda \geq \mathbf{0}$.

1.2.4 Strong Duality

For general NLPs, the duality gap may be strictly positive: the dual optimal cannot reach the primal optimal. For convex NLPs, however, the gap closes under a mild regularity condition:

Slater's Condition: There exists a point $\hat{x} \in \mathbb{R}^n$ (a Slater point) such that $g_i(\hat{x}) < 0$ strictly for all $i = 1, \dots, m$.

Strong Duality Theorem (convex NLP): If the NLP is convex (convex f , convex g_i) and Slater's condition holds, then:

1. $q^* = f^*$ — the duality gap is zero.
2. The dual optimal λ^* exists and satisfies the KKT conditions at x^* .

In other words, for convex NLPs satisfying Slater's condition, solving the dual is equivalent to solving the primal — and the dual optimal multipliers are exactly the KKT multipliers recovered by Penalty, Barrier, and Interior-Point in Lectures 16–18.

1.3 In-Class Exercises

1.3.1 Exercise 1

Same problem as Lectures 16–18, and as Lecture 19 In-Class Exercise 1

MTC Chennai is optimizing a demand-responsive feeder service on the Tambaram–Chromepet corridor. The daily operating cost depends on fleet size n (vehicles) and scheduled headway h (minutes):

$$C(n, h) = (n - 5)^2 + 2(h - 3)^2$$

The depot has a combined resource constraint — vehicles and crew shifts — that limits $n + h \leq 7$.

From Lectures 16–18, the constrained optimum is $(n^* = \frac{13}{3}, h^* = \frac{8}{3})$ with KKT multiplier $\lambda^* = \frac{4}{3}$.

Part 1 — Construct the Lagrangian and dual function.

The constraint in standard form is $g(n, h) = n + h - 7 \leq 0$. Form the Lagrangian:

$$\mathcal{L}(n, h, \lambda) = (n - 5)^2 + 2(h - 3)^2 + \lambda(n + h - 7)$$

To compute $q(\lambda) = \inf_{n, h} \mathcal{L}(n, h, \lambda)$, treat $\lambda \geq 0$ as fixed and minimize over (n, h) without any constraint:

$$\frac{\partial \mathcal{L}}{\partial n} = 2(n - 5) + \lambda = 0 \implies n^*(\lambda) = 5 - \frac{\lambda}{2}$$

$$\frac{\partial \mathcal{L}}{\partial h} = 4(h - 3) + \lambda = 0 \implies h^*(\lambda) = 3 - \frac{\lambda}{4}$$

Substituting back into $\mathcal{L}$:

$$\begin{aligned} q(\lambda) &= \left(-\frac{\lambda}{2}\right)^2 + 2\left(-\frac{\lambda}{4}\right)^2 + \lambda\left(\left(5 - \frac{\lambda}{2}\right) + \left(3 - \frac{\lambda}{4}\right) - 7\right) \\ &= \frac{\lambda^2}{4} + \frac{\lambda^2}{8} + \lambda\left(1 - \frac{3\lambda}{4}\right) = \lambda - \frac{3\lambda^2}{8} \end{aligned}$$

This is a concave quadratic in λ — confirming Property 1 (the dual function is always concave).

Part 2 — Solve the dual problem.

$$\max_{\lambda \geq 0} q(\lambda) = \lambda - \frac{3\lambda^2}{8}$$

Taking the derivative and setting it to zero:

$$\frac{dq}{d\lambda} = 1 - \frac{3\lambda}{4} = 0 \implies \lambda^* = \frac{4}{3}$$

Since q is concave, this is the global maximum. The dual optimal value is:

$$q\left(\frac{4}{3}\right) = \frac{4}{3} - \frac{3}{8} \left(\frac{4}{3}\right)^2 = \frac{4}{3} - \frac{3}{8} \cdot \frac{16}{9} = \frac{4}{3} - \frac{2}{3} = \frac{2}{3}$$

Part 3 — Verify strong duality.

The primal optimal cost is:

$$C\left(\frac{13}{3}, \frac{8}{3}\right) = \left(\frac{13}{3} - 5\right)^2 + 2\left(\frac{8}{3} - 3\right)^2 = \left(-\frac{2}{3}\right)^2 + 2\left(-\frac{1}{3}\right)^2 = \frac{4}{9} + \frac{2}{9} = \frac{2}{3}$$

$$\boxed{f^* = q^* = \frac{2}{3}} \quad \checkmark \quad (\text{zero duality gap})$$

Slater's condition is satisfied: the point $(n, h) = (1, 1)$ has $n + h = 2 < 7$, so it is strictly feasible.

Part 4 — Verify complementary slackness.

The constraint is active at the optimum: $n^* + h^* = \frac{13}{3} + \frac{8}{3} = 7$, so $g(n^*, h^*) = 0$. Complementary slackness requires $\lambda^* \cdot g(n^*, h^*) = \frac{4}{3} \cdot 0 = 0$.

Part 5 — Interpret $\lambda^* = \frac{4}{3}$.

If the combined resource budget tightens from 7 to 6 (i.e., $\epsilon = -1$), the optimal cost increases by approximately $\lambda^* \cdot 1 = \frac{4}{3}$ — an additional 1,333 per day in operating cost. Conversely, relaxing the budget from 7 to 8 saves approximately 1,333 per day. The resource is currently binding and carries a positive scarcity price.

1.3.2 Exercise 2

Same problem as Lectures 16–18 In-Class Exercise 2, and as Lecture 19's worked example

An NHA corridor study models total system travel time across two parallel links — the Mumbai–Pune Expressway (v_1 , thousand veh/hr) and the old NH48 (v_2 , thousand veh/hr):

$$T(v_1, v_2) = 3v_1^2 - 2v_1v_2 + 2v_2^2 - 14v_1 - 12v_2 + 50$$

A combined capacity constraint limits total flow: $v_1 + v_2 \leq 5$.

From Lectures 16–18, the constrained optimum is $(v_1^* = \frac{16}{7}, v_2^* = \frac{19}{7})$ with KKT multiplier $\lambda^* = \frac{40}{7}$.

Part 1 — Construct the Lagrangian and dual function.

Constraint in standard form: $g(v_1, v_2) = v_1 + v_2 - 5 \leq 0$. Lagrangian:

$$\mathcal{L}(v_1, v_2, \lambda) = 3v_1^2 - 2v_1v_2 + 2v_2^2 - 14v_1 - 12v_2 + 50 + \lambda(v_1 + v_2 - 5)$$

Minimizing over (v_1, v_2) for fixed $\lambda \geq 0$:

$$\frac{\partial \mathcal{L}}{\partial v_1} = 6v_1 - 2v_2 - 14 + \lambda = 0$$

$$\frac{\partial \mathcal{L}}{\partial v_2} = -2v_1 + 4v_2 - 12 + \lambda = 0$$

Solving this 2×2 linear system (multiply first equation by 2, add to second):

$$10v_1 - 40 + 3\lambda = 0 \implies v_1^*(\lambda) = 4 - \frac{3\lambda}{10}$$

$$v_2^*(\lambda) = \frac{12 - \lambda - 2v_1^*(\lambda)}{4} \cdot \frac{4}{4} = 5 - \frac{2\lambda}{5}$$

Substituting back, the dual function is:

$$q(\lambda) = 4\lambda - \frac{7\lambda^2}{20} - 8$$

This is a concave quadratic — confirming the dual function is always concave.

Part 2 — Solve the dual problem.

$$\frac{dq}{d\lambda} = 4 - \frac{7\lambda}{10} = 0 \implies \lambda^* = \frac{40}{7}$$

$$q\left(\frac{40}{7}\right) = 4 \cdot \frac{40}{7} - \frac{7}{20} \cdot \left(\frac{40}{7}\right)^2 - 8 = \frac{160}{7} - \frac{160}{7} \cdot \frac{1}{2} - 8 = \frac{80}{7} - 8 = \frac{24}{7}$$

Part 3 — Verify strong duality.

The primal optimal value is:

$$T\left(\frac{16}{7}, \frac{19}{7}\right) = 3\left(\frac{16}{7}\right)^2 - 2 \cdot \frac{16}{7} \cdot \frac{19}{7} + 2\left(\frac{19}{7}\right)^2 - 14 \cdot \frac{16}{7} - 12 \cdot \frac{19}{7} + 50 = \frac{24}{7}$$

$$\boxed{f^* = q^* = \frac{24}{7}} \quad \checkmark$$

Part 4 — Interpret $\lambda^* = \frac{40}{7} \approx 5.71$.

If the combined corridor capacity were reduced from 5,000 to 4,000 vehicles/hour (a tightening of 1 unit), total system travel time would increase by approximately $\frac{40}{7}$ thousand vehicle-hours — a substantial congestion cost. Conversely, a capacity expansion by 1,000 veh/hr (e.g., adding a third lane) saves approximately $\frac{40}{7}$ thousand vehicle-hours. NHAH can use this shadow price to evaluate whether the annualized cost of expansion is justified by the congestion savings.

1.4 Take-Away Exercises

1.4.1 Exercise 1

Same problem as Lectures 16–18, and as Lecture 19 In-Class Exercise 2

Delhivery is minimizing the daily operating cost at its Chennai cross-dock facility. The cost depends on daily throughput λ (tonnes/day) and staffing level s (workers):

$$C(\lambda, s) = 2\lambda^2 - 4\lambda s + 3s^2 - 12\lambda - 6s + 50$$

A combined budget and shift constraint limits $\lambda + 2s \leq 14$.

From Lectures 16–18, the constrained optimum is $(\lambda^* = \frac{116}{19}, s^* = \frac{75}{19})$ with KKT multiplier $\mu^* = \frac{64}{19}$.

1. Form the Lagrangian $\mathcal{L}(\lambda, s, \mu)$ and minimize over (λ, s) for fixed $\mu \geq 0$ by solving $\nabla_{\lambda, s} \mathcal{L} = \mathbf{0}$. Express $\lambda^*(\mu)$ and $s^*(\mu)$ as explicit functions of μ .
2. Substitute into $\mathcal{L}$ to obtain the dual function $q(\mu)$. Verify that:

$$q(\mu) = -\frac{19\mu^2}{8} + 16\mu - 49$$

and confirm this is concave.

3. Solve the dual problem $\max_{\mu \geq 0} q(\mu)$ analytically to recover $\mu^* = \frac{64}{19}$ and $q(\mu^*) = -\frac{419}{19}$.
4. Verify strong duality by computing $C(\frac{116}{19}, \frac{75}{19})$ and confirming it equals $q(\mu^*)$. Check Slater's condition: is there a strictly feasible point?
5. Interpret $\mu^* = \frac{64}{19}$ for the Delhivery operations manager. If the combined budget constraint tightens from 14 to 13 units, by how much does the optimal cost increase? What does this tell the manager about the value of adding one extra worker-shift to the budget?

1.4.2 Exercise 2

Same problem as Lectures 16–18, and as Lecture 19 Take-Away Exercise 1

BMTC is optimizing its demand-responsive feeder service in Bengaluru. The total daily operating cost depends on fleet size n (vehicles) and headway h (minutes):

$$C(n, h) = 2n^2 + 3h^2 + 2nh - 20n - 30h + 100$$

A depot capacity constraint limits $n \leq 2$, i.e., $g(n, h) = n - 2 \leq 0$.

From Lectures 16–18, the constrained optimum is $(n^* = 2, h^* = \frac{13}{3})$ with KKT multiplier $\nu^* = \frac{10}{3}$.

1. Form the Lagrangian $\mathcal{L}(n, h, \nu)$. Minimize over (n, h) by solving $\nabla_{n, h} \mathcal{L} = \mathbf{0}$ for fixed $\nu \geq 0$. Note that only the $\partial/\partial n$ stationarity condition involves ν — explain why.
2. Derive the dual function and verify that:

$$q(\nu) = -\frac{3\nu^2}{20} + \nu + 10$$

Solve the dual problem $\max_{\nu \geq 0} q(\nu)$ analytically to recover $\nu^* = \frac{10}{3}$ and $q(\nu^*) = \frac{35}{3}$.

3. Verify strong duality by computing $C(2, \frac{13}{3})$ and confirming it equals $q(\nu^*)$.
 4. Implement the dual in Python. Plot $q(\nu)$ for $\nu \in [0, 8]$ using `numpy` and `matplotlib`. Mark ν^* and $q(\nu^*)$ on the plot. On the same figure, mark the three multiplier estimates recovered by Penalty (L16), Barrier (L17), and Interior-Point (L18) as vertical lines, and compare them to $\nu^* = \frac{10}{3}$ on the dual curve.
 5. Write a brief managerial note (3–4 sentences) for the BMTC fleet planning team. Address: (a) what $\nu^* = \frac{10}{3}$ means in operational terms — if BMTC could add one extra vehicle to the depot, how much would daily operating cost fall? (b) is this saving sufficient to justify procuring an additional vehicle at a daily lease cost of 4,000?
 6. *Connecting to Lecture 19.* Lecture 19 asked you to derive $\nu^*(b)$ in closed form for this exact problem by substituting the active constraint directly into C — a *primal* derivation; the result is $\nu^*(b) = 10 - \frac{10}{3}b$. Differentiate your dual function $q(\nu)$ a second way: treat b as a parameter in the constraint $g(n, h) = n - b \leq 0$, re-derive $q(\nu; b)$, solve $\max_{\nu} q(\nu; b)$ to get $\nu^*(b)$, and confirm it matches Lecture 19’s primal result exactly. What does this agreement tell you about the relationship between the primal sensitivity derivation (Lecture 19) and the dual derivation (this lecture)?
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1.5 Further Reading

- Nocedal, J. and Wright, S.J. (2006). *Numerical Optimization* (2nd ed.). Springer — Chapter 12 (theory of constrained optimization: Lagrangian duality, KKT conditions, constraint qualifications).
- Bazaraa, M.S., Sherali, H.D., and Shetty, C.M. (2006). *Nonlinear Programming: Theory and Algorithms* (3rd ed.). Wiley — Chapter 6 (duality and saddle-point optimality conditions).
- Boyd, S. and Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press — Chapter 5 (Lagrangian duality: dual function, weak and strong duality, geometric interpretation, Slater’s condition). Freely available at web.stanford.edu/~boyd/cvxbook.
- Bertsimas, D. and Tsitsiklis, J.N. (1997). *Introduction to Linear Optimization* — Chapter 4 (LP duality as a special case of the NLP Lagrangian dual framework).
- SciPy documentation: `scipy.optimize.minimize` — docs.scipy.org